

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)		
Student Name:	SRIMATHI . B	Reg. No.:	192511448
Section / Batch:	II nd yr - BE - CSE - (AI)	Date:	

Code of Conduct

I, Srinathi . B (192511448) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Srinathi . B

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
GRAND TOTAL:			50 Marks

Short Answer

Name : B. S. Mathi

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Course code : ITA0502

Course name : Computer vision

②. Computer vision:

⇒ Computer vision is a part of Artificial intelligence that enables computers to capture, process, analyze and understand images and videos. It uses image processing and machine learning techniques to identify objects, recognize patterns and make decision automatically.

a. Main objectives:

- ⇒ To acquire the image from the real world.
- ⇒ To analyze and understand the visual data.
- ⇒ Extract useful information from images.
- ⇒ Recognize and classify objects.
- ⇒ To support ^{intelligent} decision making.

b. Two applications:

Face recognition
Self driving cars.

The key features of computer vision.

⇒ Computer vision extracts meaningful information from images using advanced algorithms. It supports

Objects, detection and recognition, tracking and immediately

a. Feature extraction:

⇒ Feature extraction is the process and identify important images characteristics such as edges, corners, textures and shapes.

b. Object recognition:

Object recognition identifies classic objects in an image improving the accuracy and intelligent of computer vision systems.

3. Different levels of image processing:

Computer vision uses three levels of image processing.

- * Low-Level - image enhancement and noise removal
- * Mid-Level - image segmentation and feature extraction.
- * High-Level - Object recognition and scene understanding.

a. Low-Level image processing:

It improves image quality by reducing noise and enhancing ~~phase~~ brightness or contrast.

b. Difference:

Mid-Level	High-Level
Extracts features and segments of objects	→ Interprets image content and makes decision

Digital image representation:

→ A digital image consists of pixels arranged in rows and columns.

⇒ Each pixel stores intensity and color information.

a. Pixel:

⇒ A pixel is the smallest element of a digital image that represents one point of color or intensity.

b. Effects of Resolution:

Higher resolution provides more pixels, resulting in clearer and sharper image.

Image sensing and Acquisition:

Image sensing and acquisition converts real image world scenes into digital images using image sensor and image devices.

a. Role of image sensor:

It converts light into electrical signals which are transformed into digital images.

b. Two application devices:

Digital camera

Scanners.

6. Sampling and Quantization:

Sampling and Quantization converts an analog image into a digital image.

a. Sampling image:

Sampling image converts a continuous image into discrete pixels.

b. Process of quantization image:

Quantization assigns intensity values to pixels allowing the image to be digitally.

7. Image formation model:

The image information model explains light from a scene is captured by a camera to produce an image.

a. Perspective projection:

It projects a 3D scene onto a 2D image plane making distant objects appear smaller.

b. Role of illumination:

Illumination provides the light necessary to capture clear and accurate images.

Processing vs Computer vision

Image processing improves image quality while computer vision interprets image content for intelligent decision making.

a. Image Processing:

It performs operation such as filtering, enhancement, and restoration of images.

b. Two difference.

Image Processing enhance images.

→ Computer vision understand the images.

Image processing outputs an improved image.

→ Computer vision outputs information or decision.

9. Application of computer vision:

Computer vision is widely used in healthcare, transportation, agriculture, robotics and security.

a. Healthcare:

It helps detect diseases from X-rays, CT scans and MRI images.

b. Autonomous vehicle:

It detects roads, traffic signals, pedestrians and obstacles for safe driving.

⑩. Digital image fundamentals:

Digital image fundamentals such as pixels, sampling and resolution are essential for accurate image analytics.

a. Why are pixels important?

=> Pixels are the basic building blocks of digital image and store color or intensity information.

b. Effect of sampling and resolution:

Higher sampling and resolution produce more detailed images, improving image analytic accuracy.